

Matrix functions and stability

# Degree coverage with seven matrix multiplications

**Difficulty:** challenging **Importance:** interesting to the community**Status:** Solved

**Rating rationale:** Historical ratings for the original conjecture: challenging because numerical evidence for coefficient-space coverage needs exact algebraic justification; community impact comes from multiplication budgets for practical matrix-function evaluation.

## Resolution — negative, 17 September 2026

**Author:** Marcus Webb, The University of Manchester. Developed with substantial ChatGPT assistance; two separate Codex agents independently reviewed the proof and recomputed its exact certificates. This is informal AI-agent review, not external human peer review or formal verification.

The conjectured equality is false. [Theorem 1 of the complete proof](#) establishes

$$\mathbb{C}[x]_{\leq 44} \subseteq \overline{\mathcal{P}_7}^Z.$$

The construction first retains a suitable quadruple of polynomials in the joint closure of four-product computations (Lemma 2). Three more products give a 45-parameter coefficient map whose exact Jacobian determinant is 256. The argument uses the original complex field and all coefficients through degree 128; high-degree coefficients vanish in the limit rather than being discarded.

Corollary 3, using Jarlebring–Lorentzon’s dimension theorem, gives  $44 \leq d_7 \leq 47$  for the maximum in the retained question. Its exact new value remains undetermined, but this does not prevent a complete negative answer to the specific equality  $d_7 = 42$ . No exact representation of every degree-44 polynomial, real Euclidean density, or numerical stability is asserted.

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## Verified partial result — 2026-09-11

**Author:** Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the stated partial results.

A fixed seven-product scheme has a full-rank complex coefficient map, certified by a nonzero exact integer Jacobian minor. Its image contains a nonempty Zariski-open subset of  $\mathbb{C}[x]_{\leq 42}$  and is Euclidean dense there.

This earlier result did not prove an upper bound excluding degree 43 and above, so at the time it left the maximal-degree equality open. It asserted neither exact representation of every polynomial nor real Euclidean dense coverage. The later degree-44 construction above now refutes that equality; the degree-42 theorem remains valid.

**Primary reference:** [complete authored PDF](#), [standalone TeX](#), [Theorem 1 and equations \(1\)-\(2\)](#). [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

## Problem statement

Start with the scalar polynomials  $1, x \in \mathbb{C}[x]$ . Allow arbitrarily many complex linear combinations of already computed polynomials and at most seven multiplications of two already computed polynomials. Let  $\mathcal{P}_7 \subseteq \mathbb{C}[x]_{\leq 128}$  be the set of possible outputs.

Identify a polynomial with its coefficient vector in  $\mathbb{C}^{129}$  and let  $\overline{\mathcal{P}_7}^Z$  be its Zariski closure: the common zero set of all polynomial equations in the coefficients that vanish throughout  $\mathcal{P}_7$ . Is

$$\max\{d \in \{0, \dots, 128\} : \mathbb{C}[x]_{\leq d} \subseteq \overline{\mathcal{P}_7}^Z\} = 42?$$

The target concerns closure and the complex coefficient field. Exact representability of every polynomial is a stronger assertion.

## Why it matters

For a dense input matrix, matrix products dominate the cost of polynomial evaluation. The conjecture determines how much polynomial degree a budget of seven products can cover, providing a theoretical benchmark for matrix-function evaluation schemes.

## References

1. E. Jarlebring and G. Lorentzon, *The polynomial set associated with a fixed number of matrix-matrix multiplications*, arXiv:2504.01500v3 (13 August 2025), §2.2 (closure convention), §5.4, Conjecture 13. [Primary text](#).
2. J. Sastre, J. Ibáñez, J. M. Alonso, and E. Defez, *Beyond Paterson–Stockmeyer: Advancing matrix polynomial computation*, WSEAS Transactions on Mathematics 24 (2025), 684–693, §4 (a five-product construction). [Primary paper](#), [DOI](#).
3. J. M. Alonso, J. Sastre, J. Ibáñez, and E. Defez, *A systematic framework for stable and cost-efficient matrix polynomial evaluation*, arXiv:2603.23143v1 (24 March 2026), abstract and conclusions. [Primary text](#).

## Status check — 2026-09-08

The latest arXiv source remains v3 and labels the displayed equality a conjecture supported by computations. The later papers give constructions and stability procedures without establishing the seven-product degree-coverage claim. Searches included the exact title with 2026, *Jarlebring Lorentzon conjecture seven multiplications 42*, and *Sastre polynomial evaluation 2026*. No resolution was located. A [3 September 2026 author talk](#) also describes higher-cost degree coverage as conjectural. Six-product real/complex formulations have a separate source ambiguity and are deliberately not included here.

## Audit — 2026-09-10

Rechecked [Conjecture 13](#) and [Remark 14](#): complex closure coverage at seven products is still conjectural, with concrete matrix-exponential applications. Title and seven-product searches found no exact proof. The [June 2026 degree-eight construction](#) solves a different budget/degree problem.